# Vigneshwaran C. J.

**AI Engineer & Full-Stack Software Developer**

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## PROFESSIONAL SUMMARY

AI Engineer and full-stack developer experienced in production-oriented AI applications, conversational commerce platforms, automation workflows, and cloud-deployed web products. Hands-on work spans Python, Node.js, TypeScript, React, FastAPI, PostgreSQL, Redis, Docker, Kubernetes, and REST APIs. Experienced in LLM integration, multi-agent workflows, OCR, payment and logistics integrations, SaaS architecture, usage tracking, and computational biology applications. Current work at Ceiyone Tech Works includes **AIORA**, a 13-microservice WhatsApp commerce platform. Selected products include PocketPet, Syncly, and GlycanBench.

## WORK EXPERIENCE

### Junior Software Developer — Ceiyone Tech Works Pvt. Ltd.

Dec 2025 – Present

Full-time

Designing and maintaining AIORA, a multi-tenant WhatsApp commerce platform spanning 13 microservices for bookings, e-commerce, document validation, payments, logistics, and AI interactions. Work covers Node.js/TypeScript and Python APIs, concurrent messaging sessions, Redis queues, OCR and GPT-4o Vision, PostgreSQL usage tracking, Razorpay and Shiprocket integrations, n8n workflows, and production deployment on Ubuntu VPS and Railway.

**Key deliverables:** Bullion Handler, Bullion Listener, Splendor, Oblion, meridian-api, Meridian, polaris-ui, AIORA E-Commerce

- WhatsApp automation for customer communication and onboarding
- REST API design with FastAPI and Express.js
- PostgreSQL and MongoDB management and query optimisation
- Multi-tenant platform design (shared WhatsApp numbers, multi-agent routing)
- Multi-environment VPS deployment (dev/SIT/prod) with systemd, PM2, nginx and isolated Postgres/Redis
- IAM, Secrets Manager, CloudWatch configuration on AWS
- Mentoring interns (code review, architecture guidance)

### AI Engineer Intern — Ceiyone Tech Works Pvt. Ltd.

Oct 2025 – Dec 2025

Internship

Built AI-powered automation solutions, OCR-based document systems, LinkedIn automation tools, and multi-agent LLM systems. Integrated AI services, external APIs, and workflow automation platforms.

**Key deliverables:** AI Weather Agent, LinkedIn Automation Platform, LinkedIn Content Creation Agent, SEO DeepAgent, Multi-Agent Financial AI, Business Card OCR System

### Research Intern — Manipal Institute of Technology

Jun 2024 – Jul 2024

Internship

Built QSPR models for Posaconazole solubility in SEDDS formulations using RDKit descriptors, VIF-based feature reduction, and regression model comparison; reported $R^2$=0.926.

**Key deliverables:** Development of QSPR Models for Self-Emulsifying Drug Delivery System (SEDDS) of Posaconazole

### Tutor — Kamakshi Tuition Centre

Jan 2023 – Jun 2023

Part-time

Taught mathematics to 9th grade students and Python programming to 12th grade students at Kamakshi Tuition Centre, Mugalivakkam Arasa Maram.

## EDUCATION

### Master of Technology in Big Data Biology

Aug 2023 – Jun 2025

SASTRA Deemed University · Thanjavur, Tamil Nadu, India · Grade: 79.67%

### Bachelor of Technology

Aug 2019 – Apr 2023

Prathyusha Engineering College · Tiruvallur, Tamil Nadu, India · Grade: 84.4%

## PUBLICATIONS & RESEARCH

### GlycanBench: a unified resource for working with glycans

Research Paper · Submitted — Frontiers in Systems Biology · Systems Computational Biology Lab, SASTRA Deemed University · 2026

Unified glycoinformatics web server and Python package with 20 tools. Includes an MPNN trained on 1,344 curated glycans after correcting 12 contradictory labels, on-demand conformers, GLYSUM alignment, network visualisat...

### Explainable Machine Learning-Based Approach to Developing Potent EGFR Inhibitors for Ovarian Cancer

Thesis · M.Tech Thesis · SASTRA Deemed University · 2025

M.Tech thesis developing an explainable ML framework for EGFR inhibitor discovery in ovarian cancer. More than 1,800 RDKit and Mordred descriptors were reduced through LASSO selection. Six classifiers were tuned using Gr...

### Prediction of lignocellulosic components and fermentable sugars for bioethanol production by machine learning approach

UG Project · B.Tech Main Project · SASTRA Deemed University · 2024

ML models (Decision Tree, Random Forest, LGBM) trained on 20 lignocellulosic biomass samples pretreated with nanotechnology and ionic liquids to predict biomass composition and ethanol yield. BFuel web platform for datas...

### Suitability Studies Of Fruits Of West Indian elm Plant (Guazuma ulmifolia) As Feedstock For Biofuel production

UG Project · B.Tech Mini Project · Prathyusha Engineering College · 2023

Investigated Guazuma ulmifolia fruits as biofuel feedstock via petroleum ether solvent extraction, emulsion and acrolein lipid assays, and H2SO4 acid hydrolysis for fermentable sugar testing. No lipids detected; carbohyd...

## KEY INDEPENDENT PROJECTS

### Syncly

Developer Networking Platform

Developer-to-developer networking platform built solo end to end. Members pick a mentorship role (mentor, learner, both), swipe a paginated feed of developers they have not met (Vibe to connect, Ghost to pass); the recipient replies Link or Noped, and a mutual Link unlocks real-time 1:1 ch...

**Stack:** React 19, Vite, Redux Toolkit, React Router 7, Tailwind CSS 4, daisyUI, Axios, Node.js, Express 5, MongoDB, Mongoose, Socket.IO, JWT, bcrypt, AWS SES, node-cron, AWS EC2

### GlycanBench: a unified resource for working with glycans

Glycoinformatics Web Server + Python Package · SASTRA University

Full-stack glycoinformatics platform and installable Python package built with Ashok Palaniappan at SASTRA's Systems Computational Biology Lab. React 19 and FastAPI expose 20 tools for glycan creation, SNFG 2D and on-demand 3D conformers, format conversion, molecular descriptors, fingerpri...

**Stack:** Python, FastAPI, Pydantic, React 19, TypeScript, Vite, Tailwind CSS 4, PyTorch, PyTorch Geometric, RDKit, glycowork, glypy, BioPython, scikit-learn, SciPy, LangChain, Groq, 3Dmol.js, Cytoscape.js, Chart.js

### Explainable Machine Learning-Based Approach to Developing Potent EGFR Inhibitors for Ovarian Cancer

M.Tech Thesis · Explainable AI · Drug Discovery

M.Tech thesis developing an explainable ML framework for EGFR inhibitor discovery in ovarian cancer. More than 1,800 RDKit and Mordred descriptors were reduced through LASSO selection. Six classifiers were tuned using GridSearchCV and four-fold cross-validation. Random Forest reached 93% a...

**Stack:** Python, Scikit-Learn, RDKit, Mordred, SHAP, Gradient Boosting, Random Forest, ChEMBL

### Prediction of lignocellulosic components and fermentable sugars for bioethanol production by machine learning approach

ML for Green Energy · Biomass Pretreatment

This study explores efficient bioethanol production from waste biomass using an integrated approach of nanotechnology, green pretreatment methods, and machine learning. Twenty different lignocellulosic biomass samples (mainly fruit peels and agro-wastes) were pretreated using physical, che...

**Stack:** Python, Scikit-Learn, Random Forest, LGBM, Decision Tree, Web Development

### QSPR for Posaconazole SEDDS

Drug Solubility Prediction · Manipal Institute of Technology

Developed QSPR models for Posaconazole solubility in Self-Emulsifying Drug Delivery Systems at Manipal Institute of Technology under Dr. Neha Sureshchandra Gandhi. Generated molecular descriptors with RDKit, reduced multicollinearity through VIF analysis, compared regression models, and re...

**Stack:** Python, RDKit, Scikit-Learn, Random Forest, PLS, Linear Regression, SVM, KNN, Mordred

### PocketPet

Desktop Companion, AI Browser Agent + Download Website

One cross-platform product comprising a Tauri 2/Rust desktop companion and its React 19/TypeScript download website. The desktop app provides JavaScript/SVG pets, games, customisation, break reminders, and a streamed task agent that searches, reads pages, and controls a dedicated Chromium ...

**Stack:** Rust, Tauri, JavaScript, React.js, TypeScript, Vite, Tailwind CSS, Tokio, Win32, Chrome DevTools Protocol, Agentic AI, GitHub API, GitHub Actions, Vercel

## CONTACT


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## TECHNICAL SKILLS

**Programming Languages**

Python · R · SQL · JavaScript · TypeScript · Rust

**Data Science & ML**

Pandas · NumPy · Scikit-Learn · PyTorch · TensorFlow · SHAP

**Frontend Development**

React.js · Next.js · Redux · HTML5 · CSS3 · Tailwind CSS · Bootstrap

**Backend Development**

Node.js · Express.js · FastAPI · Flask · Streamlit · Socket.IO · BullMQ · Mongoose · Tauri

**Databases**

MongoDB · PostgreSQL · DynamoDB · Oracle · Redis

**Cloud & DevOps**

AWS EC2 · AWS SES · Google Cloud · Firebase · Docker · PM2 · systemd · nginx · Render · Railway · GitHub Actions · Kubernetes · Minikube

**AI & LLMs**

OpenAI GPT-4o · LangChain · LangGraph · Groq · Agentic AI · Gemini · OCR · Prompt Engineering · Context Engineering

**Bioinformatics & Research**

RDKit · Mordred · glycowork · PyTorch Geometric · BioPython · Glycomics · Cytoscape · Galaxy

**Tools**

Git · GitHub · n8n · Postman · Playwright · Jupyter Notebook · Linux · Vim · Arduino · Photoshop · macOS · Puppeteer · Apify · Razorpay · Shiprocket · Webhooks · Redis Streams

## CERTIFICATIONS

**Namaste ReactJS**
NamasteDev · Akshay Saini

**Namaste Node.js**
NamasteDev · Akshay Saini

## RESEARCH INTERESTS

Graph Neural Networks, Computational Glycobiology, Agentic AI Systems, Explainable Machine Learning, Drug Discovery via ML